

Weibel Introduction to Homological Algebra Solutions

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This is a solution set to the first few problems in Weibel's "Introduction to Homological Algebra". As this is for fun, solutions are brief and generally aim to give the main ideas. I have not and do not plan to skip any problem.

Chain Complexes

1.1.1

It is a quick verification that composing two maps gives the 0 map ($4 \cdot 4 = 16 \equiv 0 \pmod{8}$). Now for $n \geq 1$ the kernel of each map is $2\mathbb{Z}$ and the image into each C_n is $4\mathbb{Z}$ so the homology $H_n(C)$ is $\mathbb{Z}/2$. For $n = 0$ the kernel of each map is $\mathbb{Z}$ and the image into it is $\mathbb{Z}/4$ so the homology is $\mathbb{Z}/4$. For everything else the kernel and homology are trivial.

1.1.2

We only need to consider a small section of the diagram.

$$\begin{array}{ccccc} C_{n+1} & \longrightarrow & C_n & \longrightarrow & C_{n-1} \\ \downarrow u_{n+1} & & \downarrow u_n & & \downarrow u_{n-1} \\ D_{n+1} & \longrightarrow & D_n & \longrightarrow & D_{n-1} \end{array}$$

We claim that u_n sends boundaries to boundaries and cycles to cycles. Note that the image of a boundary is a boundary since $u_n \circ d = d \circ u_{n+1}$. Furthermore the image of a cycle is a cycle since $u_{n-1} \circ d = d \circ u_n$.

1.1.3

Note that the composition of two maps is 0 since we have $C_n \rightarrow B_{n-1} \subset C_{n-1} \rightarrow B_{n-2}$ however B_{n-1} and B_{n-2} do not intersect in C_{n-1} . Now suppose we have a chain sequence of vector spaces

$$\cdots \rightarrow C_{n+1} \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \rightarrow \cdots$$

Then write:

$$0 \rightarrow \ker(d_n) \rightarrow C_n \rightarrow \operatorname{im}(d_n) \rightarrow 0 \quad 0 \rightarrow \operatorname{im}(d_{n+1}) \rightarrow \ker(d_n) \rightarrow H_n(C) \rightarrow 0$$

Note that vector spaces are projective, therefore they split in exact sequences. We can therefore write $C_n = \ker(d_n) \oplus \operatorname{im}(d_n)$ and $\ker(d_n) = \operatorname{im}(d_{n+1}) \oplus H_n(C)$. Putting this together we obtain $C_n = \operatorname{im}(d_n) \oplus \operatorname{im}(d_{n+1}) \oplus H_n(C)$.

1.1.4

Consider the chain complex

$$\cdots \rightarrow \text{Hom}_R(A, C_{n+1}) \rightarrow \text{Hom}_R(A, C_n) \rightarrow \text{Hom}_R(A, C_{n-1}) \rightarrow \cdots$$

Where the boundary maps are $\phi \mapsto d \circ \phi$. Then the composition of two boundary maps has $\phi \mapsto d \circ d \circ \phi = 0 \circ \phi = 0$ as desired.

Now take $A = Z_n$. Suppose that $H_n(C) \neq 0$. Then $Z_n \neq B_n$. Consider the inclusion map $\phi : Z_n \subseteq C_n$ which is an element of $\text{Hom}_R(A, C_n)$. This map is a boundary of $\text{Hom}_R(A, C_n)$ since it is 0 when postcomposed with d . Furthermore it cannot be the image of a map in $\psi \in \text{Hom}_R(A, C_{n+1})$ since the image of such a $d \circ \psi$ is strictly contained in B_n .

Now consider the converse assertion. We have an exact sequence

$$0 \rightarrow C_{n+1} \rightarrow C_n \rightarrow C_{n-1} \rightarrow 0$$

which will not generally be exact after we apply $\text{Hom}_R(A, C)$ since this functor is not left exact. For instance take

$$0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$$

and note that $Z_n = \mathbb{Z}$. Now note that

$$\text{Hom}(\mathbb{Z}, \mathbb{Z}) \cong \mathbb{Z} \quad \text{Hom}(\mathbb{Z}, \mathbb{Q}) \cong \mathbb{Q} \quad \text{Hom}(\mathbb{Z}, \mathbb{Q}/\mathbb{Z}) = 0$$

Therefore we obtain the sequence

$$0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow 0$$

which is nonsensical since $\mathbb{Z}$ is not isomorphic to $\mathbb{Q}$.

1.1.5

Note that (1) and (2) are equivalent by the definition of exactness. Furthermore the condition of the map $0 \rightarrow C$ being a quasi-isomorphism is by definition equivalent to all the $H_n(C)$ being 0.

1.1.6

We will first address $H_0(C)$. Note the the kernel of the map $C_0 \rightarrow C_{-1} = 0$ is C_0 . The map $d : C_1 \rightarrow C_0$ map a linear combination of edges $e_{ij} : v_i \rightarrow v_j$ to $ke_{ij} \mapsto k(v_i - v_j)$. This will exactly induce the relation $v_i = v_j$ in the quotient space. Therefore if the graph is connected all the edges will be "equal" to eachother in the quotient and the space will have dimension 1.

Now we will address $H_1(C)$. The image of $0 = C_2 \rightarrow C_1$ is 0 so we are asked to find the dimension of the kernel of the map $C_1 \rightarrow C_0$. Note that the matrix has V rows and E columns, and has an image of dimension 1. Therefore by rank nullity theorem the dimension of the kernel is $V - E - 1$. This corresponds to the number of circuits since each circuit essentially corresponds to a linearly independent element of the null space. IE if we have a circuit

$$v_1 \rightarrow v_2 \rightarrow \cdots \rightarrow v_n \rightarrow v_1$$

then the element $e_{12} + e_{23} + \cdots + e_{n-1,n} + e_{n1}$ is an element of the null space.

1.1.7

We can label the vertices $0, 1, 2, 3$ (I will not draw a picture since the choice is irrelevant). Each face is given by removing one of the vertices, so we call the orientation of the i -th face (which is removing the i -th vertex) $(-1)^i$ with respect to the induced orientation of the larger simplex. More generally, given an edge of a simplex, its orientation is defined with respect to which vertices we removed and the order we removed them. To determine the orientation of a general "sub-simplex", first write all the ommitted vertices *in the order they were omitted*, then write all the vertices it contains according to the ambient order of the larger simplex. The sign of this permutation is the orientation of the edge. This show that $d^2 = 0$ since for $n > 1$ we have that S_n has an equal number of odd and even permutations.

Now we write the maps. I will denote $s_{a_1, \dots, a_n}$ as the sub simplex spanned by the $a_1, \dots, a_n$ vertices. Then a direct computation gives the matrices:

$$\begin{array}{c|cccc} & s_{012} & s_{013} & s_{023} & s_{123} \\ \hline s_{01} & 1 & 1 & 0 & 0 \\ s_{02} & -1 & 0 & 1 & 0 \\ s_{03} & 0 & -1 & -1 & 0 \\ s_{12} & 1 & 0 & 0 & 1 \\ s_{13} & 0 & 1 & 0 & -1 \\ s_{23} & 0 & 0 & 1 & 1 \end{array} \quad \begin{array}{c|cccccc} & s_{01} & s_{02} & s_{03} & s_{12} & s_{13} & s_{23} \\ \hline s_0 & -1 & -1 & -1 & 0 & 0 & 0 \\ s_1 & 1 & 0 & 0 & -1 & -1 & -1 \\ s_2 & 0 & 1 & 0 & 1 & 0 & 0 \\ s_3 & 0 & 0 & 1 & 0 & 1 & 1 \end{array}$$

1.2.1

We claim that in an abelian category maps $A \rightarrow C$ and $B \rightarrow D$ induce maps of the biproducts $A \times B \rightarrow C \times D$. Furthermore this preserves exactness. This is because a map out of the biproduct is specified by maps out of each summand. Now compose by the natural injections

$$A \rightarrow C \rightarrow_{l_C} C \times D \quad B \rightarrow D \rightarrow_{l_D} C \times D$$

which specifies a map $A \times B \rightarrow C \times D$ by the other universal property of the biproduct.

Now consider exact sequences of the form

$$0 \rightarrow A \rightarrow C \rightarrow E \rightarrow 0 \quad 0 \rightarrow B \rightarrow D \rightarrow F \rightarrow 0$$

and construct the sequeunce:

$$0 \rightarrow A \times B \rightarrow C \times D \rightarrow E \times F \rightarrow 0$$

We may test for exactness by applying the Hom functors.

$$\text{Hom}(A \times B, P) \rightarrow \text{Hom}(C \times D, P) \rightarrow \text{Hom}(E \times F, P) \rightarrow 0$$

$$\text{Hom}(P, E \times F) \rightarrow \text{Hom}(P, C \times D) \rightarrow \text{Hom}(P, A \times B) \rightarrow 0$$

and all these Homs decompose as direct product by the universal property. We have essentially reduced to problem to the right exactness of biproduct on abelian groups. This is by a diagram chase.

$$G_1 \times G_2 \rightarrow G_3 \times G_4 \rightarrow G_5 \times G_6 \rightarrow 0$$

every element of the kernel is by given by some $g_3 \times g_4$ with $g_3 \mapsto 0$ and $g_4 \mapsto 0$. This gives g_1 and g_2 so that $g_1 \times g_2 \mapsto g_3 \times g_4$ by exactness of the two sequences.

Then we have:

$$0 \rightarrow \text{im}(d_{n-1}^\alpha) \rightarrow \ker(d_n^\alpha) \rightarrow H_n(A_\alpha) \rightarrow 0$$

$$0 \rightarrow \text{im}(d_{n-1}^\beta) \rightarrow \ker(d_n^\beta) \rightarrow H_n(A_\beta) \rightarrow 0$$

At this point we note that homology is an abelian category, so we can take the biproduct of the exact sequences.

$$0 \rightarrow \text{im}(d_{n-1}^\alpha) \times \text{im}(d_{n-1}^\beta) \rightarrow \ker(d_n^\alpha) \times \ker(d_n^\beta) \rightarrow H_n(A_\alpha) \times H_n(A_\beta) \rightarrow 0$$

1.2.2

We will consider the map $h : A \rightarrow B$.

First of all, every kernel $K \rightarrow_k A \rightarrow_h B$ is mono. This is by the universal property. Pick $f, g : C \rightarrow K$. Then both f and g correspond uniquely maps $\tilde{f} : C \rightarrow A$ and $\tilde{g} : C \rightarrow A$. These correspond to $k \circ f$ and $k \circ g$. Therefore k is left cancellative.

Similarly cokernels $A \rightarrow_h B \rightarrow_c C$ are epis. Choose $f, g : C \rightarrow D$. Then both f and g correspond uniquely to maps $B \rightarrow D$ which corresponds to $f \circ c$ and $g \circ c$. Therefore this map is right cancellative and is an epi.

Now suppose that $f : A \rightarrow B$ and using the abelian condition we construct the kernel of its cokernel

$$\begin{array}{ccccc} & \text{ker coker } f & & & \\ & \downarrow u & \searrow h & & \\ A & \xrightarrow{f} & B & \xrightarrow{g} & \text{coker}(f) \end{array}$$

Note that $g \circ f = 0$ and therefore since $\text{ker coker } f$ is universal with respect to this property we get a map $u : \text{ker coker } f \rightarrow A$ with $u \circ f = h$. We that u is always an isomorphism when f is monic. Similarly if we construct the maps

$$\begin{array}{ccccc} & & & \text{coker ker } f & \\ & & \nearrow h & \downarrow u & \\ \text{ker } f & \xrightarrow{g} & A & \xrightarrow{f} & B \end{array}$$

We see again that u must be an isomorphism when f is epi.

1.2.3

Consider the diagram of maps:

$$\begin{array}{ccccccc} & \text{ker}(f_{n+1}) & & \text{ker}(f_n) & & \text{ker}(f_{n-1}) & \\ & \downarrow \iota_{n+1} & & \downarrow \iota_n & & \downarrow \iota_{n-1} & \\ \cdots & \longrightarrow & A_{n+1} & \xrightarrow{d_{n+1}} & A_n & \xrightarrow{d_n} & A_{n-1} \longrightarrow \cdots \\ & \downarrow f_{n+1} & & \downarrow f_n & & \downarrow f_{n-1} & \\ \cdots & \longrightarrow & B_{n+1} & \xrightarrow{d_{n+1}} & B_n & \xrightarrow{d_n} & B_{n-1} \longrightarrow \cdots \\ & \downarrow \pi_{n+1} & & \downarrow \pi_n & & \downarrow \pi_{n-1} & \\ & \text{coker}(f_{n+1}) & & \text{coker}(f_n) & & \text{coker}(f_{n-1}) & \end{array}$$

We need to define the maps on the top and the bottom. Consider the composition $d_{n+1} \circ \iota_{n+1} : \text{ker}(f_{n+1}) \rightarrow A_n$. By commutativity of the diagram this map is killed by precomposition by f_n . Therefore by the universal property of the kernel it gives a map $d_{n+1} : \text{ker}(f_{n+1}) \rightarrow \text{ker}(f_n)$. The composition of two of these maps $d_n \circ d_{n+1}$ must be zero by left cancellativity after postcomposing with ι_{n-1} (which gives the 0 map by commutativity).

Now we will define the maps on the bottom. The map $d_{n+1} : B_{n+1} \rightarrow B_n$ is killed by precomposition by f_{n+1} and therefore gives a map $\text{coker}(f_{n+1}) \rightarrow B_n$ which extends to a map $\text{coker}(f_{n+1}) \rightarrow \text{coker}(f_n)$ after postcomposing with π_n . The composition of two of these maps $d_n \circ d_{n+1}$ must be zero by right cancellativity after precomposition with π_{n+1} (which gives 0 by commutativity).

1.2.4

The statement that

$$0 \rightarrow A \rightarrow_a B \rightarrow_b C \rightarrow 0$$

Is exactly the statement that

- (a) a is monic.
- (b) b is an epi.
- (c) $\ker a = \operatorname{coker} b$.

However we showed in 1.2.3 that all these conditions are checked piecewise. Therefore this is equivalent to

$$0 \rightarrow A_n \rightarrow B_n \rightarrow C_n \rightarrow 0$$

being exact for all n .

1.2.5

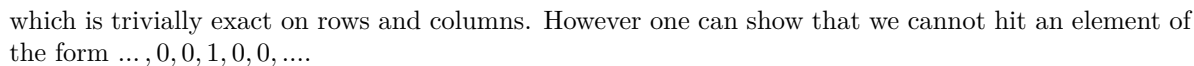
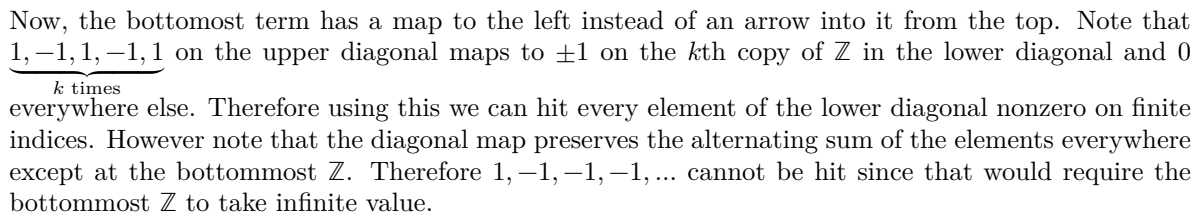
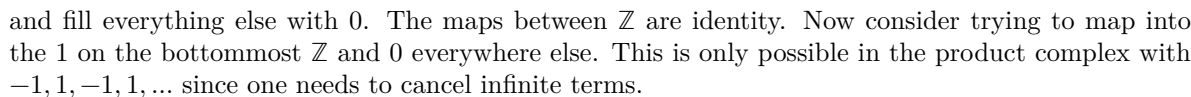
Suppose that the columns are exact (ie $\ker d_k^v = \operatorname{im} d_{k+1}^v$) and consider a section of the diagram:

$$\begin{array}{ccccc}
 & & C_{p-1,q+1} & & \\
 & & \downarrow d^v & & \\
 & C_{p-1,q} & \xleftarrow{d^h} & C_{p,q} & \\
 & \downarrow d^v & & \downarrow d^v & \\
 C_{p-1,q-1} & \xleftarrow{d^h} & C_{p,q-1} & \xleftarrow{d^h} & C_{p+1,q-1} \\
 & & \downarrow d^v & & \downarrow 0 \\
 & & C_{p,q-2} & \xleftarrow{0} & 0
 \end{array}$$

By boundedness we may assume that $C_{p,q-1}$ has the maximum value of $p - q$ along the line $p + q = n$ (indicated by the 0). We will focus only on 3 diagonals so relabel the diagram like this:

$$\begin{array}{ccccc}
 & & A_1 & & \\
 & & \downarrow d^v & & \\
 & B_1 & \xleftarrow{d^h} & A_0 & \\
 & \downarrow d^v & & \downarrow d^v & \\
 C_1 & \xleftarrow{d^h} & B_0 & \xleftarrow{d^h} & A_{-1} \\
 & & \downarrow d^v & & \downarrow 0 \\
 & & C_0 & \xleftarrow{0} & 0
 \end{array}$$

So that A, B, C are the n th, $n - 1$ th and $n - 2$ th elements of the total chain complex. Suppose that we have an element $b = b_0 \times b_1 \times \dots$ of the kernel. We claim that we can lift this to some $a_0 \times a_1 \times \dots$ mapping into it. Since there is no horizontal contribution to C_0 we have exactness at $A_0 \rightarrow B_0 \rightarrow C_0$ and we can determine a_0 so that $d(a_0)|_{B_0} = b_0$. Therefore $b - da_0$ has no B_0 component, and clearly represents the same element of $\operatorname{coker}(d : A \rightarrow B)$. We replace b with $b - da_0$ and induct on the minimal value of n such that B_n has a nonzero component of our element b . By boundedness this induction will terminate and we have shown that d represents the 0 element in $\operatorname{coker}(d : A \rightarrow B)$.



1.2.7

Recall that exactness of a complex means exactness on each column. If we draw it out we see that we must show is exactness at $0 \rightarrow Z_k \rightarrow C_k \rightarrow B_{k-1} \rightarrow 0$. However by defintion Z_k is the kernel of the map $C_k \rightarrow C_{-1}$ and B_{k-1} is the image of the map.

For the second we see that we must show exactness of the following sequence:

$$0 \rightarrow H_k \rightarrow C_k/B_k \rightarrow Z_{k-1} \rightarrow H_{k-1} \rightarrow 0$$

Since $H_k = Z_k/B_k$ we obtain the first injection whose image is Z_k/B_k . This is by definition the kernel of the differential map. The image of the differential map is B_{k-1} inside Z_{k-1} and the cokernel map gives a surjective map to H_{k-1} which is exact.

1.2.8

We define the mapping cone D as:

$$D_n = B_{n-1} \oplus C_n \quad d = \begin{bmatrix} d_B & 0 \\ f & -d_C \end{bmatrix}$$

We need to show that there is an exact sequence

$$0 \rightarrow C_n \rightarrow B_{n-1} \oplus C_n \rightarrow B_{n-1} \rightarrow 0$$

which is clear. We can note that D can

1.3.1

Note that if a complex is exact then the homology is 0. If we constuct the long exact sequence of cohomology then since the terms cycle between A , B , and C parts, if two are exact then the homologies of the others are sandwiched between 0s. This forces them to be 0 by exactness as desired.

1.3.2

We can consider the rows as chain complex and the diagram as an exact sequence of chain complexes. Then all the claims follows from 1.3.1.

1.3.3

First we have a claim:

Claim 1.0.1 (Strong 4 Lemma). Consider the diagram with exact rows

$$\begin{array}{ccccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & C & \xrightarrow{h} & D \\ \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \downarrow \delta \\ A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' & \xrightarrow{h'} & D' \end{array}$$

if α is epic and δ is a monic then $g(\ker \beta) = \ker \gamma$ and $g'^{-1}(\text{im } \gamma) = \text{im } \beta$.

The proof is a diagram chase, I have written some diagrams below:

$$\begin{array}{ccccccc} a_7 & & b_4 & \longrightarrow & c_0 & \longrightarrow & 0_3 \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ a'_6 & \longrightarrow & b'_5 & \longrightarrow & 0_1 & \longrightarrow & 0_2 \end{array} \qquad \begin{array}{ccccccc} b_4 & \longrightarrow & c_1 & \longrightarrow & 0_3 \\ & & \downarrow & & \downarrow \\ b'_0 & \longrightarrow & c'_1 & \longrightarrow & 0_2 \end{array}$$

The order in which the elements were constructed is marked via subscript. For the first claim:

- Start with $c_0 \in \ker \gamma$, by commuting $\delta h(c_0) = h' \gamma(c_0) = 0$. This gives us $0_1, 0_2, 0_3$.
- By exactness we find b_4 with $g(b_4) = c_0$. We let $b'_5 = \beta(b_4)$, which is killed by g' by commutativity.
- By exactness get a'_6 mapping to b'_5 and then use surjectivity to lift to a_7 .
- By construction, $b_4 - f(a_7)$ is killed by β and is mapped to c_0 by g as desired.

Now we perform the second diagram chase for the second claim:

- Start with $b'_0 \in g'^{-1}(\text{im } \gamma)$ and let $c'_1 = g'(b'_0)$. By the assumption we have c_1 with $\gamma(c_1) = c'_1$.
- By exactness $h'(c'_1) = 0$, therefore by injectivity of δ and commutativity of the diagram $h(c_1) = 0$
- By exactness we find b_4 with $g(b_4) = c_1$.
- Note that $b'_0 - \beta(b_4)$ is killed by g' , so there is some $a'_5 \in A'$ with $f'(a'_5) = b'_0 - \beta(b_4)$.
- Using surjectivity of α we find $a_6 \in A$ with $\alpha(a_6) = a'_5$. They by construction $b_4 + f(a_6)$ is sent to b'_0 by β as desired.

As a corollary, if β is monic then $g(0) = \ker \gamma$ so $\ker \gamma = 0$ and γ is monic. Dually if γ is epic then $g^{-1}(C') = \text{im } \beta$ so $\text{im } \beta = B'$ and β is epic. The five lemma follows from applying this result to the 2 subdiagrams in the five lemma. Giving injectivity and surjectivity.

1.3.4

The maps $H^n(C) \rightarrow H^n(B(-1))$ are trivial and factor through 0. Therefore splitting up the long exact sequence by 0s along these maps gives the result.

1.3.5

The map $f : A \rightarrow B$ splits into two short exact sequences:

$$0 \rightarrow \ker f \rightarrow A \xrightarrow{f} \text{im } f \rightarrow 0 \quad 0 \rightarrow \text{im } f \rightarrow B \rightarrow \text{coker } f \rightarrow 0$$

Since $\ker f$ and $\text{coker } f$ are acyclic, when we take the long exact sequence of cohomology, we have segments of the form

$$0 \rightarrow H_n(A) \xrightarrow{f} H_n(\text{im } f) \rightarrow 0 \quad 0 \rightarrow H_n(\text{im } f) \rightarrow H_n(B) \rightarrow 0$$

we see that f is an isomorphism $H_n(A) \xrightarrow{\sim} H_n(\text{im } f) \cong H_n(B)$ and we are done.

1.3.6

The maps $\text{Tot}(A) \rightarrow \text{Tot}(B) \rightarrow \text{Tot}(C)$ are the standard maps

$$\bigoplus_{p+q=n} A_{pq} \rightarrow \bigoplus_{p+q=n} B_{pq} \rightarrow \bigoplus_{p+q=n} C_{pq}$$

We can individually analyze each component of the map:

$$0 \rightarrow A_{pq} \rightarrow B_{pq} \rightarrow C_{pq} \rightarrow 0$$

We now check what needs to be checked.

- The first map is injective since the image of $a \in \text{Tot}(A)$ being zero implies that every component a_{pq} is zero by the definition of the maps.

- The second map is surjective since for any $c \in \text{Tot}(C)$, we can find some b_{pq} mapping to each component c_{pq} . These then assemble to a $b \in \text{Tot}(C)$ mapping to c .
- The composition is 0 since it is an assembly of a bunch of compositions which are 0.
- An element of the kernel of the second map, is made up of a bunch of b_{pq} going to 0. Therefore we can find a_{pq} hitting them and assemble into an $a \in \text{Tot}(A)$ hitting b .

We conclude that if $\text{Tot}(C)$ is acyclic then $\text{Tot}(A) \rightarrow \text{Tot}(B)$ is a quasi-isomorphism. This is because the long exact sequence splits into blocks of the form

$$0 \rightarrow H_n(\text{Tot}(A)) \rightarrow H_n(\text{Tot}(B)) \rightarrow 0$$

implying the isomorphism.

1.4.1

- (a) We perform this by induction. The base case is a map $0 \rightarrow 0$ (by boundedness), which is clearly split. Suppose that we have constructed the splitting map up to a certain point:

$$C_{n+1} \xrightarrow{d} C_n \xleftarrow{s_{n-1}} C_{n-1}$$

We have the diagram:

$$\begin{array}{ccc} & C_{n+1} \oplus C_{n-1} & \\ s_n \nearrow & \downarrow (d, s_{n-1}) & \\ C_n & \xrightarrow{\quad} & C_n \end{array}$$

Where the vertical map is specified by the maps $d : C_{n+1} \rightarrow C_n$ and $s_{n-1} : C_{n-1} \rightarrow C_n$. The horizontal map is the identity map. We claim that the vertical map is surjective, at which point we obtain s_n via the projectivity of C_n . For any element $x \in C_n$, we let $x' = ds_{n-1}(x)$. By the splitting we have $d(x') = d(x)$ so $d(x - x') = 0$. Therefore by exactness there is some $y \in C_{n+1}$ with $d(y) = x - x'$. Therefore $(s_{n-1}(x), y)$ maps to x via (d, s_{n-1}) .

- (b) We use the fact that free abelian groups are exactly free $\mathbb{Z}$ -modules, and $\mathbb{Z}$ is a PID. We have the usual exact sequence:

$$0 \xrightarrow{i} Z_n \hookrightarrow C_n \xrightarrow{d} B_{n-1} \longrightarrow 0$$

however by exactness we can write this as:

$$0 \xrightarrow{i} B_n \hookrightarrow C_n \xrightarrow{d} B_{n-1} \longrightarrow 0$$

Since $\mathbb{Z}$ is a PID we know that submodules of a free module are free. Therefore B_{n-1} is free, hence projective. By projectivity the sequence is split exact and we obtain a map $t_{n-1} : B_{n-1} \rightarrow C_n$ with $d \circ t_{n-1} = \text{id}_{B_{n-1}}$. We also get $r_n : C_n \rightarrow B_n$ such that $r_n \circ i = \text{id}$. We form the composition

$$C_n \xrightarrow{r_n} B_n \xrightarrow{t_n} C_{n+1}$$

and claim that $s_n = t_n \circ r_n$. In particular

$$dsd = dt_n r_n d = \text{id}_{B_{n-1}} r_n d = d$$

1.4.2

$\Rightarrow$: Suppose that there is an R -module decomposition $C_n \cong Z_n \oplus B'_n$ and $Z_n \cong B_n \oplus H'_n$. We have the maps:

$$C_n \xrightarrow{\pi} B'_n \xrightarrow{d} B_{n-1}$$

where π_1 is the standard projection onto B'_n . Since the second map is injective and surjective, we can define an inverse $r_{n-1} : B_{n-1} \rightarrow B'_n$. Now the composition

$$C_{n-1} \xrightarrow{\pi_2} B_{n-1} \xrightarrow{r_{n-1}} B'_n$$

splits the exact sequence. Here π_2 is the standard projection map.

$\Leftarrow$: We have the decomposition:

$$C_n = \ker(d) \oplus \operatorname{im}(s_{n-1} \circ d)$$

which we identify with Z_n and B'_n respectively. We furthermore have the decomposition:

$$Z_n = \operatorname{im}(d \circ s_n) \oplus \ker(d \circ s_n)$$

This works since $s_{n-1} \circ d$, as well as $d \circ s_n$, are both their own square, hence projections in the sense that matters.

1.4.3

$\Rightarrow$ Recall from the exposition at the beginning of the sections that $ds + sd$ is id when the sequence is split exact. Then the identity is by definition null-homotopic.

$\Leftarrow$ Suppose that the identity is null-homotopic so that $ds + sd = \operatorname{id}$. Then taking d on both sides yields $dsd = d$ as desired.

1.4.4

It turns out that the converse is actually true.

$\Rightarrow$: Suppose that the complex C is split, so we have s with $d = dsd$. Recall that the kernel of the map $ds + sd$ can be identified with $H_*(C)$. Let

$$\pi : C_* \rightarrow H_*(C) \quad \pi = \operatorname{id} - (ds + sd)$$

be the projection onto this kernel. The inclusion $\ker(ds + sd) \rightarrow C_*$ induces a map $\iota : H_*(C) \rightarrow C_*$. Now $\pi \circ \iota$ is exactly the identity map on $H_*(C)$ by construction. Moreover $\iota \circ \pi$ is exactly $\operatorname{id} - (ds + sd)$ so $\operatorname{id} - \iota \circ \pi = ds + sd$ showing that π and ι induce a chain homotopy equivalence as desired.

$\Leftarrow$: Now suppose that there is a chain homotopy equivalence $f : C_* \rightarrow H_*(C)$ and $g : H_*(C) \rightarrow C_*$. We have that

$$\operatorname{id} - gf = s_1 d + ds_1 \quad \operatorname{id} - fg = s_2 d + ds$$

However since the differential map on the homology is 0, we see that $s_1 d + ds_1 = 0$. Taking d on the right of the second equation we see that

$$d - dfg = ds_2 d$$

Note that since d commutes with f we have that $dfg = fdg = 0$ since the differential is 0 on $H_*(C)$. Therefore

$$d = ds_2 d$$

as desired.

1.4.5

- (a) To show that it is an equivalence relation, we note that it is clearly reflexive and symmetric. Suppose that $f \simeq g \simeq h$. Then

$$f - g = s_1 d + d s_2 \quad g - h = s_2 d + d s_2$$

therefore

$$f - h = (s_1 + s_2) d + d(s_1 + s_2)$$

as desired. We see that $\text{Hom}_K(C, D)$ is an abelian group since there is a nice addition operation that we prove in (3). Addition is associative, there is an identity map, and the negative of a map is the inverse.

- (b) Draw the diagram:

$$\begin{array}{ccc} B & & E \\ \downarrow u & & \uparrow v \\ C & \xrightleftharpoons[g]{f} & D \end{array}$$

Since f and g are chain homotopic we have $f - g = sd + ds$. Therefore

$$vfu - vgu = v(sd + ds)u = (vsu)d + d(vs u)$$

since d must commute with these maps. This demonstrates that maps in $\mathbf{K}$ are associative under function composition, therefore $\mathbf{K}$ is indeed a category.

- (c) By assumption we have that

$$f_i - g_i = s_i d + d s_i$$

Therefore

$$(f_0 + f_1) - (g_0 + g_1) = (s_0 + s_1) d + d(s_0 + s_1)$$

so the addition operation respects the equivalence relation. This makes the category additive since

- The maps form an abelian group by (1).
- It is true of $\mathbf{Ch}$ that addition distributes over function composition. Since every map in $\mathbf{K}$ has a representative in $\mathbf{Ch}$, functoriality means that $\mathbf{K}$ inherits this property:

$$\bar{f}(\bar{a} + \bar{b})\bar{g} = \overline{f(a + b)g} = \overline{fag + fbg} = \bar{f}\bar{a}\bar{g} + \bar{f}\bar{b}\bar{g}$$

- The biproduct in $\mathbf{K}$ is the image of the biproduct in $\mathbf{Ch}$.

The induced map on the Hom sets are a morphism of abelian groups so the functor $\mathbf{Ch} \rightarrow \mathbf{K}$ is additive.

- (d) It turns out not to be an abelian category. Indeed, $\mathbf{K}$ is triangulated. In any triangulated category every monomorphism and every epimorphism splits: if $A \xrightarrow{u} B \rightarrow C \rightarrow A[1]$ is a triangle and u is monic, then $C[-1] \rightarrow A \xrightarrow{u} B$ has composite zero, hence $C[-1] \rightarrow A$ is zero, so the triangle splits; the epi case is dual.

Suppose $\mathbf{K}$ were abelian. Let $X = \mathbb{Z}[0]$ and let $f = 2: X \rightarrow X$. In an abelian category, f factors as

$$X \twoheadrightarrow \text{im } f \hookrightarrow X.$$

Since epis and monos split in $\mathbf{K}$, there is some $g \in \text{End}_{\mathbf{K}}(X)$ with

$$fgf = f.$$

But

$$\text{End}_{\mathbf{K}}(\mathbb{Z}[0]) \simeq \text{End}_{\mathbf{Ch}}(\mathbb{Z}) \simeq \mathbb{Z}.$$

Thus g is multiplication by some $n \in \mathbb{Z}$, and $fgf = f$ says

$$2n2 = 2,$$

i.e. $4n = 2$, impossible. Hence $\mathbf{K}$ is not abelian.

1.5.1

We define

$$s = \begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix}$$

and when $f = \text{id}$ we have

$$d = \begin{bmatrix} -d_C & 0 \\ -1 & d_C \end{bmatrix}$$

Note that

$$dsd = \begin{bmatrix} -d_C & 0 \\ -1 & d_C \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -d_C & 0 \\ -1 & d_C \end{bmatrix} = \begin{bmatrix} -d_C & 0 \\ -1 & d_C \end{bmatrix}$$

after an explicit computation and an application of $d_C^2 = 0$.

1.5.2

We want to understand when f extends to a map $(-s, f) : \text{cone}(C) \rightarrow D$. The thing to check is that this map commutes with the differential map:

$$\begin{array}{ccc} \text{cone}(C)_n & \xrightarrow{d_{\text{cone}(C)}} & \text{cone}(C)_{n-1} \\ \downarrow f & & \downarrow f \\ D_n & \xrightarrow{d_D} & D_{n-1} \end{array}$$

We have

$$d \circ (-s, f) = d_D \circ [-s, f] = [-d_D s, d_D f]$$

and

$$(-s, f) \circ d = [-s, f] \begin{bmatrix} -d_C & 0 \\ -1 & d_C \end{bmatrix} = [sd_C - f, fd_C]$$

since $d_D f = fd_C$ (in order for f to be a well defined complex) the second coordinate of the matrices are the same. To match the first coordinate we exactly need s to have the property that

$$f = sd + ds$$

which is exactly the statement that the map f is null-homotopic as desired.

1.5.3

As before, we need the differential map to commute with (f, s, g) :

$$d \circ (f, s, g) = (df, ds, dg)$$

$$(f, s, g) \circ d = (f, s, g) \begin{bmatrix} d_C & 1 & 0 \\ 0 & -d_C & 0 \\ 0 & -1 & d_C \end{bmatrix} = (fd, f - sd - g, gd)$$

Since $df = fd$ and $dg = gd$ all the coordinates are equal except the second, which requires:

$$ds + sd = f - g$$

which is exactly the statement that f and g are chain homotopic.

1.5.4

It is clear that $\beta\alpha = \text{id}_C$ since:

$$\alpha(c) = (0, 0, c) \quad \beta(0, 0, c) = c$$

Now compute

$$\alpha\beta(b, b', c) = \alpha(f(b) + c) = (0, 0, f(b) + c)$$

which has the matrix form:

$$\alpha\beta = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ f & 0 & 1 \end{bmatrix}$$

And furthermore

$$\begin{aligned} sd + ds &= \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} d_B & 1 & 0 \\ 0 & -d_B & 0 \\ 0 & -f & d_C \end{bmatrix} + \begin{bmatrix} d_B & 1 & 0 \\ 0 & -d_B & 0 \\ 0 & -f & d_C \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 \\ d_B & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ -d_B & 0 & 0 \\ -f & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -f & 0 & 0 \end{bmatrix} \end{aligned}$$

We see directly from this that $\text{id} - \alpha\beta = sd + ds$, proving that s is a chain homotopy between the identity and $\alpha\beta$. Now by definition α is a chain homotopy equivalence between C and $\text{cyl}(f)$

1.5.5

In this case, it is clear that $D = \text{coker}(f)$ on the module level. IE D is the chain complex with $\text{coker}(f)$ at degree 0 and zeros elsewhere. The map $\varphi : \text{cone}(f) \rightarrow D$ is

$$\begin{array}{ccccccc} \text{cone}(f) : & 0 & \longrightarrow & B & \xrightarrow{-f} & C & \longrightarrow 0 \\ & & & & & \downarrow \pi & \\ D : & 0 & \longrightarrow & 0 & \longrightarrow & \text{coker}(f) & \longrightarrow 0 \end{array}$$

where π is the standard quotient map. Firstly, we may assume $-f$ is injective. If there is a split exact sequence, then $-f$ must be injective by defintion. Conversely, since D is clearly acyclic, $-f$ must be injective so that $\text{cone}(f)$ is also acyclic. For this to be a chain homotopy equivalence, we need a map $\psi : \text{coker}(f) \rightarrow C$ satisfying two conditions:

- $\text{id} - \pi\psi = ds + sd$ however since the differential map on D is 0, this requires $\pi\psi = \text{id}$. Therefore ψ must be a retract of π . This implies that the exact sequence

$$0 \rightarrow B \xrightarrow{-f} C \xrightarrow{\pi} \text{coker}(f) \rightarrow 0$$

is split by ψ . In fact, ψ splits the exact sequence if and only if it satisfies this condition. Therefore this condition is equivalent to the sequence being split exact.

- $\text{id} - \psi\pi = ds + sd$. Here s is a map $C \rightarrow B$. Since the first condition is already equivalent to the sequence

$$0 \rightarrow B \xrightarrow{-f} C \xrightarrow{\pi} \text{coker}(f) \rightarrow 0$$

being split exact. We only need to show that this sequence being split exact allows us to find $s : C \rightarrow B$ with $\text{id} - \psi\pi = sd + ds$. By substituting in the explicit form of d we need:

$$\text{id} - \psi\pi = -fs$$

For every element $x \in C$, we may write it uniquely as $x = a + \psi\pi(x)$ where $a := x - \psi\pi(x)$. Since $\pi\psi = \text{id}_{\text{coker}(f)}$ we see that $(\psi\pi)^2$ and $\pi(a) = 0$. We write

$$s(x) := (-f)^{-1}(a)$$

which is well defined since $\pi(a)$ being 0 implies that it is uniquely the image of something in B . Under this definition a direct computation yields that

$$x - \psi\pi(x) = (ds + sd)(x) = -fs(x) = a$$

as desired.

1.5.6

From the remark given before this problem, we have the commuting square:

$$\begin{array}{ccc} H_n(\text{cone}(f)) & \xrightarrow{\partial} & H_{n-1}(B) \\ \downarrow \cong & & \parallel \\ H_n(D) & \xrightarrow{\partial} & H_{n-1}(B) \end{array}$$

which tells us that $\partial : H_n(D) \rightarrow H_{n-1}(B)$ is exactly the map:

$$H_n(D) \cong H_n(\text{cone}(f)) \xrightarrow{\partial} H_{n-1}(B)$$

Now from Lemma 1.5.7 we have the commuting square

$$\begin{array}{ccc} H_n(\text{cone}(f)) & \xrightarrow{-\partial} & H_{n-1}(B) \\ \parallel & & \parallel \\ H_n(\text{cone}(f)) & \xrightarrow{\delta} & H_n(B[-1]) \end{array}$$

which tells us that $\delta : H_n(\text{cone}(f)) \rightarrow H_{n-1}(B)$ is exactly the map:

$$H_n(\text{cone}(f)) \xrightarrow{-\delta_*} H_n(B[-1]) = H_{n-1}(B)$$

and substituting in this definition yields that the original boundary map is

$$H_n(D) \cong H_n(\text{cone}(f)) \xrightarrow{-\delta_*} H_n(B[-1]) = H_{n-1}(B)$$

1.5.7

Now for the second part of the problem consider the big diagram:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & B[-1] & \xrightarrow{f} & C[-1] & \xrightarrow{g} & D[-1] & \longrightarrow & 0 \\ & & \downarrow \psi & & \downarrow \text{id} & & & & \\ 0 & \longrightarrow & D & \xrightarrow{\iota} & \text{cone}(g) & \xrightarrow{\delta} & C[-1] & \longrightarrow & 0 \\ & & \downarrow & & \downarrow \text{id} & & & & \\ 0 & \longrightarrow & C & \longrightarrow & \text{cyl}(g) & \longrightarrow & \text{cone}(g) & \longrightarrow & 0 \end{array}$$

which creates homology long exact sequences

$$\begin{array}{ccccccc}
\cdots & \rightarrow & H_{n+1}(C[-1]) & \rightarrow & H_{n+1}(D[-1]) & \xrightarrow{\partial} & H_n(B[-1]) \rightarrow H_n(C[-1]) \rightarrow \cdots \\
& & \downarrow \text{id} & & \downarrow \text{id} & & \downarrow \psi_* \\
\cdots & \rightarrow & H_{n+1}(C[-1]) & \xrightarrow{\partial} & H_n(D) & \xrightarrow{\iota_*} & H_n(\text{cone}(g)) \rightarrow H_n(C[-1]) \xrightarrow{\partial} \cdots \\
& & \downarrow \text{id} & & \downarrow \cong & & \downarrow \text{id} \\
\cdots & \xrightarrow{\partial} & H_n(C) & \longrightarrow & H_n(\text{cyl}(g)) & \longrightarrow & H_n(\text{cone}(g)) \xrightarrow{\partial} H_{n-1}(C) \longrightarrow \cdots
\end{array}$$

By the five lemma the map $\psi_* : H_n(B[-1]) \rightarrow H_n(\text{cone}(g))$ is a quasi-isomorphism. By the commutativity of the homology diagram at the top middle square,

$$\partial : H_n(D) \rightarrow H_{n-1}(B)$$

is the map

$$H_n(D) \xrightarrow{\iota_*} H_n(\text{cone}(g)) \cong H_{n-1}(B),$$

where ι_* is the map induced by the inclusion $D \xrightarrow{\iota} \text{cone}(g)$. (Credit to Eric on stackexchange for the diagrams).

1.5.8

We may apply the diagram in 1.5.8 to the exact sequence

$$0 \rightarrow C \xrightarrow{v} \text{cone}(f) \xrightarrow{\delta} B[-1] \rightarrow 0$$

yielding the diagram:

$$\begin{array}{ccccccc}
0 & \longrightarrow & \text{cone}(f) & \longrightarrow & \text{cone}(v) & \xrightarrow{\delta} & C[-1] \longrightarrow 0 \\
& & \downarrow \alpha & & \parallel & & \\
0 & \longrightarrow & C & \longrightarrow & \text{cyl}(v) & \longrightarrow & \text{cone}(v) \longrightarrow 0 \\
& & \parallel & & \downarrow \beta & & \downarrow \varphi \\
0 & \longrightarrow & C & \xrightarrow{v} & \text{cone}(f) & \xrightarrow{\delta} & B[-1] \longrightarrow 0
\end{array}$$

and by 1.5.8 $\varphi : \text{cone}(v) \rightarrow B[-1]$ is a quasi-isomorphism as desired.

1.5.9

The map $\ker(f)[-1] \rightarrow \text{cone}(f)$ is given by the following:

$$\ker(f) \rightarrow \text{cone}(f) \quad b \mapsto (b, 0)$$

This is well defined since

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} [-d_B] \quad \begin{bmatrix} -d_B & 0 \\ -f & d_C \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

act the same on any element of B_{n-1} which is killed by f . Since the map is injective we have the exact sequence

$$0 \rightarrow \ker(f)[-1] \xrightarrow{\alpha} \text{cone}(f) \xrightarrow{\beta} \text{coker}(f) \rightarrow 0$$

which induces the long exact sequence

$$\cdots \xrightarrow{\partial} H_{n-1}(\ker(f)) \xrightarrow{\alpha} H_n(\text{cone}(f)) \xrightarrow{\beta} H_n(\text{coker}(f)) \xrightarrow{\partial} H_{n-2}(\ker(f)) \rightarrow \cdots$$

as desired. It turns out that $\text{coker}(f)$ is isomorphic to $\text{cone}(\iota)$ where ι is the inclusion $\text{im}(f) \rightarrow C$.

1.5.10

The matrix of the proposed splitting is:

$$\sigma = \begin{bmatrix} -s & 0 \\ -s'fs & s' \end{bmatrix}$$

This is a splitting when $d = dsd$. We can compute:

$$\begin{aligned} d\sigma d &= \begin{bmatrix} -d_C & 0 \\ -f & d_{C'} \end{bmatrix} \begin{bmatrix} -s & 0 \\ -s'fs & s' \end{bmatrix} \begin{bmatrix} -d_C & 0 \\ -f & d_{C'} \end{bmatrix} \\ &= \begin{bmatrix} -d_C s d_C & 0 \\ -f s d_C + d_{C'} s' f s d_C - d_{C'} s' f & d_{C'} s' d_{C'} \end{bmatrix} \\ &= \begin{bmatrix} -s & 0 \\ -f s d_C + d_{C'} s' f s d_C - d_{C'} s' f & s' \end{bmatrix} \end{aligned}$$

For this to be equal to σ we need to match the lower left entry, written as

$$(1 - d_{C'} s') f (1 - s d_C) = 0.$$

Let

$$p_C = 1 - s d_C, \quad q_{C'} = 1 - d_{C'} s'.$$

Then $p_C^2 = p_C$ and $q_{C'}^2 = q_{C'}$. Moreover p_C projects C onto $Z(C) = \ker d_C$, and $q_{C'}$ kills $B(C') = \operatorname{im} d_{C'}$ and induces the identity on $H(C')$. Thus

$$(1 - d_{C'} s') f (1 - s d_C) = q_{C'} f p_C = 0$$

if and only if, for every cycle $z \in Z(C)$, the cycle $f(z)$ has trivial homology class in $H(C')$. Equivalently,

$$q_{C'} f p_C = 0 \iff f_* : H(C) \longrightarrow H(C') \text{ is the zero map.}$$

1.6.1

This is a pre-sheaf since inclusion of open sets induces the backwards map corresponding to restriction of functions. It is a sheaf since by definition a function is C^∞ iff it is locally C^∞ . Being C^∞ is a local condition so this is a sheaf.

1.6.2

As before, inclusion of open sets corresponds to function restriction in a well defined way, so this is a pre-sheaf. A function is continuous iff the preimage of any open set is open. By point set topology, for a cover U_α of X , a set $f^{-1}(V)$ is open if and only if all the $f^{-1}(V) \cap U_\alpha$ are open. Therefore being continuous is a local condition, and the sheaf axiom is satisfied.

1.6.3

It is clearly true that the definition given in this exercise implies the one in Definition 1.6.6. Therefore suppose F has the property of Definition 1.6.6. For a sequence For any injective map

$$A \xrightarrow{f} B$$

we have the exact sequence:

$$0 \rightarrow A \xrightarrow{f} B \rightarrow \operatorname{coker}(f) \rightarrow 0$$

which becomes:

$$0 \rightarrow FA \xrightarrow{Ff} FB \rightarrow F\operatorname{coker}(f)$$

showing that Ff is injective, IE the functor preserves injective maps. Consider

$$0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C$$

we have by the existence and universal property of cokernels the following diagram with an exact row:

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \xrightarrow{f} & B & \xrightarrow{\pi} & \text{coker}(f) \longrightarrow 0 \\ & & & & & \searrow g & \downarrow \pi_2 \\ & & & & & & C \end{array}$$

Therefore we have the following diagram with a left exact row:

$$\begin{array}{ccccccc} 0 & \longrightarrow & FA & \xrightarrow{Ff} & FB & \xrightarrow{F\pi} & F\text{coker}(f) \\ & & & & & \searrow Fg & \downarrow F\pi_2 \\ & & & & & & FC \end{array}$$

Since π_2 was initially injective by exactness, $F\pi_2$ is injective. Therefore $\ker(F\pi) = \ker(Fg)$ and so exactness of the top row implies that

$$0 \rightarrow FA \xrightarrow{Ff} FB \xrightarrow{Fg} FC$$

is exact as desired. The proof for right exactness is similar.

Derived Functors

2.1.1

Let X, X' denote two short exact sequences and $f : X \rightarrow X'$ be a map between them:

$$\begin{array}{ccccccc} X : & 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow C \longrightarrow 0 \\ & & & \downarrow f_A & & \downarrow f_B & & \downarrow f_C \\ X' : & 0 & \longrightarrow & A' & \longrightarrow & B' & \longrightarrow C' \longrightarrow 0 \end{array}$$

For δ_i to be a natural transformation between the functor sending X to $T_i(C)$ and $T_{i-1}(A)$ respectively the following diagram must commute:

$$\begin{array}{ccc} T_i(C) & \xrightarrow{\delta_i} & T_{i-1}(A) \\ \downarrow \tilde{f}_C & & \downarrow \tilde{f}_A \\ T_i(C') & \xrightarrow{\delta_i} & T_{i-1}(A') \end{array}$$

This is exactly the second condition in definition 2.1.1 as desired.

2.1.2

Since there is only 1 nonzero term, the only condition that is not vacuously true is the first one. For the short exact sequence

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

We require the long exact sequence:

$$\cdots 0 \rightarrow T_0(A) \rightarrow T_0(B) \rightarrow T_0(C) \rightarrow 0 \cdots$$

which is exactly the statement that $T_0 = F$ is exact.

2.2.1

Suppose first that the chain complex is a split exact sequence of projective objects: We start with the following diagram

$$\begin{array}{ccccccc}
 \cdots & \longrightarrow & Y_1 & \longrightarrow & Y_0 & \longrightarrow & Y_{-1} \longrightarrow \cdots \\
 & & \downarrow g_1 & & \downarrow g_0 & & \downarrow g_{-1} \\
 \cdots & \longrightarrow & X_1 & \longrightarrow & X_0 & \longrightarrow & X_{-1} \longrightarrow \cdots \\
 & & \uparrow f_1 & & \uparrow f_0 & & \uparrow f_{-1} \\
 \cdots & \longrightarrow & P_1 & \longrightarrow & P_0 & \longrightarrow & P_{-1} \longrightarrow \cdots
 \end{array}$$

and seek to lift the chain map $f : P \rightarrow X$ to a map $h : P \rightarrow Y$ such that $f = g \circ h$. Recall for a split exact chain complex we can write

$$P_i = Z_i \oplus Z_{i-1}$$

so that the cycles are direct summands of projective modules, hence projective. Therefore define maps $z_i : Z_{i-1} \rightarrow Y_i$ using projectivity. Now define:

$$h = [dz_i \quad z_{i-1}]$$

We first must check that the chain map commutes with the differential maps. Note that

$$dh = [ddz_i \quad dz_{i-1}] = [0 \quad dz_{i-1}]$$

$$hd = [z_i \quad dz_{i-1}] \begin{bmatrix} 0 & \text{id}_{Z_{i-1}} \\ 0 & 0 \end{bmatrix} = [0 \quad dz_{i-1}]$$

so this is a well defined map of chain complexes. Now we need that $f = g \circ h$. Note that

$$g_i \circ h_i|_{Z_{i-1}} = g_i \circ z_{i-1} = f_i|_{Z_{i-1}}$$

and moreover

$$g_i \circ h_i|_{Z_i} = g_i \circ dz_i = dg_{i-1}z_i = \underbrace{df|_{Z_i}}_{\text{on } P_{i+1}} = \underbrace{f|_{Z_i}}_{\text{on } P_i}$$

after ignoring our severe abuse of notation. Since the map is specified by the map on the summands we are done.

Now we prove the other direction, that is, suppose P is a projective chain complex. We first note that every element P_i is projective by considering complexes X and Y , with

$$Y_i \twoheadrightarrow X_i$$

$$X_j = Y_j = 0 \quad \text{for all } j \neq i$$

and trying to extend a map $P \rightarrow X$ (given by a map $P_i \rightarrow X_i$) to a map $P \rightarrow Y$. Now we show the complex is split exact. Recall the exact sequence:

$$0 \rightarrow P \xrightarrow{\iota} \text{cone}(P) \xrightarrow{\delta} P[-1] \rightarrow 0$$

Since P is projective, $P[-1]$ is also projective so this exact sequence splits. In particular there is a retraction $r : \text{cone}(P) \rightarrow P$ with $r\iota = \text{id}_P$. Note that r is an extension of the identity map $P \rightarrow P$ to the cone, so by exercise 1.5.2 the identity map is null-homotopic. Therefore by exercise 1.4.3 the chain complex is split exact.

2.2.2

Suppose that I have a chain complex C , to each C_i , use the fact that $\mathcal{A}$ has enough projectives to find a surjection $g_i : P_i \rightarrow C_i$ with P_i projective: Now form the chain complex:

$$\begin{array}{ccccccc} \cdots & \longrightarrow & P_1 \oplus P_2 & \longrightarrow & P_0 \oplus P_1 & \longrightarrow & P_{-1} \oplus P_0 \longrightarrow \cdots \\ & & \downarrow f_1 & & \downarrow f_0 & & \downarrow f_{-1} \\ \cdots & \longrightarrow & C_1 & \longrightarrow & C_0 & \longrightarrow & C_{-1} \longrightarrow \cdots \end{array}$$

We define the differential maps on the top to be

$$d_i = \begin{bmatrix} 0 & 0 \\ \text{id}_{P_i} & 0 \end{bmatrix}$$

which clearly satisfies $d^2 = 0$ and makes the complex split. We define the maps on the bottom to be

$$f_i = [g_i, dg_{i+1}]$$

which are surjective since g_i are surjective. A direct computation show that

$$df = fd = [dg_{i+1} \quad 0]$$

so this defines a well defined map. Since direct sums of projective modules are projective, we have shown to top complex satisfies all the condition to be projective in **Ch** and we are done.

2.2.3

We need to extend the map via $f_0 : P_0 \rightarrow M$ and $f_{i \geq 1} : P_i \rightarrow 0$. Since all the $f_{i \geq 1}$ are mapping to 0, they are the zero map and the only trouble is finding a map $f_0 : P_0 \rightarrow M$.

2.2.4

Consider the diagram

$$\begin{array}{ccccccc} & & 0 & & 0 & & 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ \cdots & \longrightarrow & P'_{n+1} & \longrightarrow & P'_n & \longrightarrow & P'_{n-1} \longrightarrow \cdots \\ & & \downarrow & & \downarrow i_n & & \downarrow i_{n-1} \\ \cdots & \longrightarrow & P_{n+1} & \longrightarrow & P_n & \longrightarrow & P_{n-1} \longrightarrow \cdots \\ & & \downarrow & & \downarrow \pi_n & & \downarrow \pi_{n-1} \\ \cdots & \longrightarrow & P''_{n+1} & \longrightarrow & P''_n & \longrightarrow & P''_{n-1} \longrightarrow \cdots \\ & & \uparrow & & \uparrow & & \uparrow \\ & & 0 & & 0 & & 0 \end{array}$$

We will use commutativity of the diagram to figure out what the differential map is. Since $P_i = P'_i \oplus P''_i$ it suffices to determine the block "matrix" of the map. By commutativity

- $i_{n-1}d = di_n$. In matrix form this is

$$d \begin{bmatrix} \text{id}'_n \\ 0 \end{bmatrix} = \begin{bmatrix} d' \\ 0 \end{bmatrix}$$

which tells us that the first column of the matrix of d is $\begin{bmatrix} d' \\ 0 \end{bmatrix}$.

- $\pi_{n-1}d = d\pi_n$. In matrix form

$$\begin{bmatrix} 0 & \text{id}_{n-1}'' \end{bmatrix} d = \begin{bmatrix} 0 & d_n'' \end{bmatrix}$$

which tells us that the second row of the matrix is $\begin{bmatrix} 0 & d_n'' \end{bmatrix}$.

There is only the indetermined element on the top right of the matrix, which is our map $\lambda_n : P_n'' \rightarrow P_{n-1}'$.

2.3.1

The ideals of $\mathbb{Z}/m$ are (d) where d is a divisor of m . A map

$$d\mathbb{Z}/m \xrightarrow{f} \mathbb{Z}/m$$

is determined by the image of d . Since $f(d)(m/d) = f(m) = 0$ we see d is mapped to one of its multiples. The map is therefore multiplication by integer, say n . Then the map

$$\mathbb{Z}/m \xrightarrow{\times n} \mathbb{Z}/m$$

is an extension of this map.

2.3.2

Let $a \in A$ be an arbitrary nonzero element. In order to show that e_A is an injection, we need to find a map $A \rightarrow \mathbb{Q}/\mathbb{Z}$ which does not kill a . There are two cases

a is torsion: Let n be the order of a and consider the map

$$f : \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Q}/\mathbb{Z} \quad 1 \mapsto \frac{1}{n}$$

By the diagram and the fact that $\mathbb{Q}/\mathbb{Z}$ is injective:

$$\begin{array}{ccc} \mathbb{Z}/n\mathbb{Z} & \xrightarrow{1 \mapsto a} & A \\ \downarrow f & \swarrow f' & \\ \mathbb{Q}/\mathbb{Z} & & \end{array}$$

this extends to a map $f' : A \rightarrow \mathbb{Q}/\mathbb{Z}$ which does not kill a by commutativity.

a is not torsion: Consider the map

$$f : \mathbb{Z} \rightarrow \mathbb{Q}/\mathbb{Z} \quad 1 \mapsto \frac{1}{2}$$

By the diagram and the fact that $\mathbb{Q}/\mathbb{Z}$ is injective:

$$\begin{array}{ccc} \mathbb{Z} & \xrightarrow{1 \mapsto a} & A \\ \downarrow f & \swarrow f' & \\ \mathbb{Q}/\mathbb{Z} & & \end{array}$$

this extends to a map $f' : A \rightarrow \mathbb{Q}/\mathbb{Z}$ which does not kill a by commutativity.

2.3.3

Use contrapositive. We showed in exercise 2.3.2 that if A has a nonzero element then there is a map $A \rightarrow \mathbb{Q}/\mathbb{Z}$ which does not kill a . This map cannot be 0, so $\text{Hom}_{\mathbf{Ab}}(A, \mathbb{Q}/\mathbb{Z})$ is not 0.

2.3.4

Note that a chain complex is injective iff its dual is projective in the opposite category.

By exercise 2.2.1 a chain complex is injective iff its dual is a split exact of projective objects. Therefore since $\mathbf{Ch}(\mathcal{A})^{op} \cong \mathbf{Ch}(\mathcal{A}^{op})$ it is injective if it is a split exact complex of injective objects.

2.3.5

Let $m_0 \in M$ be an arbitrary nonzero element. In order to show that e_M is an injection, we need to find a map $M \rightarrow \text{Hom}_{\mathbf{Ab}}(R, \mathbb{Q}/\mathbb{Z})$ which does not kill m_0 . Considerig M as an abelian group under addition, by our solution to exercise 2.3.2 there is a map $f \in \text{Hom}_{\mathbf{Ab}}(M, \mathbb{Q}/\mathbb{Z})$ which does not kill m . By Lemma 2.3.8 there is the isomorphism

$$\text{Hom}_{\mathbf{Ab}}(M, \mathbb{Q}/\mathbb{Z}) \xrightarrow{\sim} \text{Hom}_{\mathbf{mod}-R}(M, \text{Hom}_{\mathbf{Ab}}(R, \mathbb{Q}/\mathbb{Z}))$$

which carries f to the map

$$\tilde{f}: m \mapsto (r \mapsto f(mr))$$

which has

$$\tilde{f}(m_0)(1) = f(m_0) \neq 0$$

so $\tilde{f}$ does not kill m_0 as desired.

2.3.6

Note that

$$\begin{aligned} \text{Hom}_{\mathbf{Ab}}(\mathcal{F}_x, A) &= \text{Hom}_{\mathbf{Ab}}(\varinjlim \{\mathcal{F}_U : x \in U\}, A) \\ &= \varprojlim \{\text{Hom}_{\mathbf{Ab}}(\mathcal{F}_U, A) : x \in U\} \end{aligned}$$

By definition, a sheaf homomorphism in $\text{Hom}_{\text{Sheaves}(X)}(\mathcal{F}, x_*A)$ is a system of maps on open sets which is compatible with restriction maps. This is exactly the inverse limit above. Therefore

$$\text{Hom}_{\mathbf{Ab}}(\mathcal{F}_x, A) \cong \text{Hom}_{\mathbf{Ab}}(\mathcal{F}_U, x_*A)$$

as desired. We see that x_*A is right adjoint to the *stalk* functor. I claim that the stalk functor is exact. In particular exactness of sheaves

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

means exactness on every open set

$$0 \rightarrow A|_U \rightarrow B|_U \rightarrow C|_U \rightarrow 0$$

and in particular a direct limit of exact sequences is exact:

$$0 \rightarrow A_x \rightarrow B_x \rightarrow C_x \rightarrow 0$$

so the stalk functor is exact as desired. Now by proposition 2.3.10 we see that x_*I is injective if I is an injective object. Since product fo injectives is injective we see that if A_x is an injective abelian group the sheaf

$$\prod_{x \in X} x_*(A_x)$$

is an injective object.

2.3.7

We desire to show that for $A \in \mathcal{A}^I$ and $B \in \mathcal{A}$:

$$\mathrm{Hom}_{\mathcal{A}}(A(k), B) \cong \mathrm{Hom}_{\mathcal{A}^I}(A, k_* B)$$

Consider a natural transformation η on the RHS of the equation. We can define a map

$$\mathrm{Hom}_{\mathcal{A}^I}(A, k_* B) \rightarrow \mathrm{Hom}_{\mathcal{A}}(A(k), B)$$

by associating η with the map:

$$\eta_k|_{\mathrm{id}} : A(k) \xrightarrow{\eta_k} \prod_{\mathrm{Hom}_I(k, k)} \xrightarrow{\pi_{\mathrm{id}}} B$$

injectivity: For $i \in I$ and $f : i \rightarrow k$ I claim that the composition

$$A(i) \xrightarrow{\eta_i} \prod_{\mathrm{Hom}_I(i, k)} \xrightarrow{\pi_f} B$$

is determined by $\eta_k|_{\mathrm{id}}$, at which point the entire map, and hence the natural transformation is determined, proving injectivity. Naturality requires the following square to commute.

$$\begin{array}{ccc} A(i) & \xrightarrow{\eta_i} & \prod_{\mathrm{Hom}_I(i, k)} B \\ \downarrow A(f) & & \downarrow \text{index map of } f \\ A(k) & \xrightarrow{\eta_k} & \prod_{\mathrm{Hom}_I(k, k)} B \end{array}$$

By the definition of the index map the copy of B projecting onto the $\mathrm{id} \in \mathrm{Hom}_I(k, k)$ copy of B below is the $f \in \mathrm{Hom}(i, k)$ index. Therefore by commutativity

$$\eta_i|_f = \eta_k|_{\mathrm{id}} \circ A(f)$$

as desired.

surjectivity: Motivated by our arguments, given a map $h : A(k) \rightarrow B$ and $f : i \rightarrow k$ we define η via

$$\eta_i|_f = h \circ A(f)$$

and see that for a map $h : i \rightarrow j$ the square

$$\begin{array}{ccc} A(i) & \xrightarrow{\eta_i} & \prod_{\mathrm{Hom}_I(i, k)} B \\ \downarrow A(h) & & \downarrow \text{index map of } h \\ A(j) & \xrightarrow{\eta_j} & \prod_{\mathrm{Hom}_I(j, k)} B \end{array}$$

commutes, so this is a well defined natural transformation as desired.

This proves the desired adjointness. Since the k th coordinate is exact k_* preserves injectives by 2.3.10.

Adopting the notation of the exercise statement, we must now show that the product

$$E = \prod_{k \in I} k_* A_k$$

exists in $\mathcal{A}^I$ and is injective. Completeness of $\mathcal{A}^I$ is inherited from completeness of $\mathcal{A}$, so the product exists. Since each $k_* A_k$ is injective (k_* preserves injectives) E is injective as a product of injectives. Since F injects into E , we have shown that $\mathcal{A}^I$ has enough injectives.

2.3.8

Note that $\mathcal{A}^I$ has enough projectives if and only if the opposite category has enough injectives. By exercise 2.3.7 and $(\mathcal{A}^I)^{op} \cong (A^{op})^I$, so this occurs when A^{op} is complete and has enough injectives. IE when $\mathcal{A}$ is co-complete and has enough injectives.

2.4.1

Let $P \rightarrow A$ and $Q \rightarrow B$ be projective resolutions. For any morphism $f : A \rightarrow B$, there exists a chain map lift $\tilde{f} : P \rightarrow Q$. Because F is a right exact functor, the sequence $F(P_1) \rightarrow F(P_0) \rightarrow F(A) \rightarrow 0$ is exact. This yields a canonical isomorphism $\eta_A : L_0F(A) \xrightarrow{\sim} F(A)$, where $L_0F(A) = \text{coker}(F(d_1))$. Applying F to the resolutions and passing to the 0-th homology yields the following diagram:

$$\begin{array}{ccc} L_0F(A) & \xrightarrow[\sim]{\eta_A} & F(A) \\ L_0F(f) \downarrow & & \downarrow F(f) \\ L_0F(B) & \xrightarrow[\sim]{\eta_B} & F(B) \end{array}$$

By the functoriality of F and the universal property of the cokernel, this square commutes. Therefore, under the natural identifications η_A and η_B , the induced map $L_0F(f)$ coincides exactly with $F(f)$.

2.4.2

Let C be a chain complex in $\mathcal{B}$. Write the sequence

$$C_{i+1} \rightarrow Z_i(C) \rightarrow H_i(C) \rightarrow 0$$

and apply the right exact functor F to obtain:

$$FC_{i+1} \rightarrow FZ_i(C) \rightarrow FH_i(C) \rightarrow 0$$

Since F is left exact (preserves kernels) we can commute F and Z_i to obtain

$$FC_{i+1} \rightarrow Z_i(FC) \rightarrow FH_i(C) \rightarrow 0$$

showing that $FH_i(C)$ naturally fits into the same exact sequence as $H_i(FC)$ so they are isomorphic.

2.4.3

Choose once and for all a resolution of A by F -acyclic modules C_i . Notice that M_m is exactly the kernel of m th the differential map. Therefore we can construct the diagram where everything is exact:

$$\begin{array}{ccccccc} & & 0 & & 0 & & \\ & \searrow & & \swarrow & & \searrow & \\ & & M_3 & & M_1 & & \\ & \swarrow & & \searrow & & \swarrow & \\ \dots & \longrightarrow & C_3 & \longrightarrow & C_2 & \longrightarrow & C_1 & \longrightarrow & C_0 & \longrightarrow & A & \longrightarrow & 0 \\ & & & & \nearrow & & \searrow & & \nearrow & & \searrow & & \\ & & & & M_2 & & M_0 & & & & & & \\ & & & & \nearrow & & \searrow & & \nearrow & & \searrow & & \\ & & & & 0 & & 0 & & 0 & & 0 & & \end{array}$$

We consider the exact sequence

$$0 \rightarrow M_0 \xrightarrow{j_0} C_0 \xrightarrow{p_0} A \rightarrow 0$$

and take the long exact sequence in homology. Using the fact that C_0 is acyclic we see that for $i \geq 2$ the long exact sequence splits into terms of the form

$$0 \rightarrow L_n F(A) \xrightarrow{\partial} L_{n-1} F(M_0) \rightarrow 0$$

and the $i = 1$ segment of the long exact sequence is

$$0 \rightarrow L_1 F(A) \xrightarrow{\partial} F(M_0) \xrightarrow{Fj_0} FC_0 \rightarrow 0$$

using the fact that L_0 is just the identity functor. This shows that

$$L_i F(A) \cong \begin{cases} L_{i-1} F(M_0) & i \geq 2 \\ \ker(Fj_0) & i = 1 \end{cases}$$

We repeat the same argument on the short exact sequence

$$0 \rightarrow M_m \xrightarrow{j_m} C_m \xrightarrow{p_m} M_{m-1} \rightarrow 0$$

to see that

$$L_i F(M_{m-1}) \cong \begin{cases} L_{i-1} F(M_m) & i \geq 2 \\ \ker(Fj_m) & i = 1 \end{cases}$$

so by induction:

$$L_i F(A) \cong \begin{cases} L_{i-m-1} F(M_m) & i \geq m+2 \\ \ker(Fj_m) & i = m+1 \end{cases}$$

We have proven that $L_i F(A) \cong \ker(Fj_{i-1})$. Apply the right exact functor F to our short exact sequence:

$$F(M_i) \xrightarrow{Fj_m} F(C_i) \xrightarrow{F(p_i)} F(M_{i-1}) \rightarrow 0$$

We will apply the first isomorphism theorem to p_i .

- $F(p_i)$ is surjective, and its kernel is the image of Fj_m , which is $\text{im}(Fd_{i+1})$.
- We restrict Fp_i to the subspace $\ker(Fd_i)$. Since $Fd_i = Fj_{i-1} \circ Fp_i$, this restriction maps $\ker(Fd_i)$ exactly onto $\ker(Fj_{i-1})$.

Now applying the first isomorphism theorem yields:

$$\frac{\ker(Fd_i)}{\text{im}(Fd_{i+1})} \cong \ker(Fj_{i-1}) \cong L_i F(A)$$

Therefore the homology of an F -acyclic resolution computes $L_i F(A)$.

2.4.4

Suppose inductively that $\varphi_i : T_i \rightarrow H_i$ are already defined (and making all the relevant diagrams commute) for $0 \leq i \leq n-1$. Given an $A \in \mathbf{Ch}(\mathcal{A})$ denote

$$\Gamma_A : \quad 0 \rightarrow A[1] \rightarrow \text{cone}(A[1]) \rightarrow A \rightarrow 0$$

for the cone exact sequence, and denote its connecting morphisms by

$$\sigma_A^T : T_m(A) \rightarrow T_{m-1}(A[1]), \quad \sigma_A^H : H_m(A) \rightarrow H_{m-1}(A[1]).$$

Since $H_n(\text{cone}(A[1])) = 0$ for all n we get the commutative diagram with exact rows:

$$\begin{array}{ccccccc} T_n(A) & \xrightarrow{\sigma_A^T} & T_{n-1}(A[1]) & \longrightarrow & T_{n-1}(\text{cone}(A[1])) & & \\ & & \downarrow \varphi_{n-1} & & \downarrow \varphi_{n-1} & & \\ 0 & \longrightarrow & H_n(A) & \xrightarrow{\sigma_A^H} & H_{n-1}(A[1]) & \longrightarrow & 0 \end{array}$$

A diagram chase reveals a unique map $\varphi_n(A)$ from $T_n(A)$ to $H_n(A)$ commuting with the σ 's.

To see naturality, consider a map $f : A' \rightarrow A$. We have the map of exact sequences:

$$\begin{array}{ccccccc} 0 & \longrightarrow & A'[1] & \longrightarrow & \text{cone}(A'[1]) & \longrightarrow & A' \longrightarrow 0 \\ & & \downarrow f[1] & & \downarrow \text{diag}(f[1], f) & & \downarrow f \\ 0 & \longrightarrow & A[1] & \longrightarrow & \text{cone}(A[1]) & \longrightarrow & A \longrightarrow 0 \end{array}$$

To see that φ_n commutes with f , we build the diagram:

$$\begin{array}{ccccc} T_n(A') & \xrightarrow{T_n(f)} & T_n(A) & & \\ \downarrow \varphi_n(A') & \searrow \sigma_{A'}^T & \downarrow \sigma_A^T & \swarrow & \downarrow \varphi_n(A) \\ & T_{n-1}(A'[1]) & \xrightarrow{T_{n-1}(f[1])} & T_{n-1}(A[1]) & \\ & \downarrow \varphi_{n-1} & & \downarrow \varphi_{n-1} & \\ & H_{n-1}(A'[1]) & \xrightarrow{H_{n-1}(f[1])} & H_{n-1}(A[1]) & \\ \uparrow \sigma_{A'}^H & & \uparrow \sigma_A^H & & \\ H_n(A') & \xrightarrow{H_n(f)} & H_n(A) & & \end{array}$$

We want to prove that the big quadrilateral commutes.

- The inner square commutes by induction.
- The top and bottom quadrilaterals commute by naturality of connecting homomorphisms
- The left and right quadrilaterals commute by construction of φ .
- The σ^H s are isomorphisms

Putting this all together forces the outer square to commute. Therefore ϕ_n is a natural transformation.

Finally let

$$E : \quad 0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

be a short exact sequence of complexes. The final thing is to prove that φ_n commutes with the connecting homomorphisms δ_E^H and δ_E^T . Now consider the following 3×3 diagram of short exact sequences:

$$\begin{array}{ccccccc} & 0 & & 0 & & 0 & \\ & \downarrow & & \downarrow & & \downarrow & \\ 0 & \longrightarrow & A[1] & \longrightarrow & B[1] & \longrightarrow & C[1] \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \text{cone}(A[1]) & \longrightarrow & \text{cone}(B[1]) & \longrightarrow & \text{cone}(C[1]) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & 0 & & 0 & & 0 & \end{array}$$

The columns are the cone exact sequences $\Gamma_A, \Gamma_B, \Gamma_C$, and the top row is the shifted exact sequence. From this 3×3 diagram, the connecting morphisms satisfy the usual anticommutativity relation

$$\sigma_A^T \circ \delta_E^T = -\delta_{E[1]}^T \circ \sigma_C^T, \quad (1)$$

$$\sigma_A^H \circ \delta_E^H = -\delta_{E[1]}^H \circ \sigma_C^H. \quad (2)$$

Therefore we may draw the diagram:

$$\begin{array}{ccccc}
T_n(C) & \xrightarrow{\delta_E^T} & T_{n-1}(A) & & \\
\downarrow \varphi_n(C) & \searrow \sigma_C^T & \swarrow \sigma_A^T & & \downarrow \varphi_{n-1}(A) \\
& T_{n-1}(C[1]) & \xrightarrow{\delta_E^T} & T_{n-2}(A[1]) & \\
& \downarrow \varphi_n(C[1]) & & \downarrow \varphi_{n-1}(A[1]) & \\
& H_{n-1}(C[1]) & \xrightarrow{\delta_E^H} & H_{n-2}(A[1]) & \\
& \swarrow \sigma_C^H & \searrow \sigma_A^H & & \\
H_n(C) & \xrightarrow{\delta_E^H} & H_{n-1}(A) & &
\end{array}$$

Now make the following arguments:

- The inner square commutes by induction.
- The top and bottom quadrilaterals anti-commute by what we've shown.
- The left and right quadrilaterals commute by the construction of φ .
- Both σ_C^H and σ_A^H are isomorphisms.

Putting this all together forces the outer square to commute. Thus φ_n commutes with connecting morphisms, so $\varphi : T \rightarrow H$ is a morphism of homological δ -functors. The argument for cohomology follows from dualizing this statement.

2.4.5

Suppose that T_* is a homological δ -functor such that each T_n is coeffaceable, and that S_* is another homological δ -functor. Suppose inductively that $\varphi_i : S_i \rightarrow T_i$ are already defined (and making all the relevant diagrams commute) for $0 \leq i \leq n-1$. Fix some $A \in \mathbf{Ch}(\mathcal{A})$ and find some coeffacing object P_n for T_n . Construct the exact sequence

$$\Gamma_A^n : \quad 0 \rightarrow K_n \rightarrow P_n \xrightarrow{u_n} A \rightarrow 0$$

and denote its connecting morphisms by

$$\sigma_{A_n}^S : S_m(A) \rightarrow S_{m-1}(K_n), \quad \sigma_{A_n}^T : T_m(A) \rightarrow T_{m-1}(K_n).$$

where $K_n := \ker(u_n)$. Taking the long exact sequence and draw the diagram:

$$\begin{array}{ccccccc}
S_n(A) & \xrightarrow{\sigma_{A_n}^T} & S_{n-1}(A[1]) & \longrightarrow & S_{n-1}(\text{cone}(A[1])) & & \\
& & \downarrow \varphi_{n-1} & & \downarrow \varphi_{n-1} & & \\
0 \rightarrow T_n(A) & \xrightarrow{\sigma_{A_n}^T} & T_{n-1}(A[1]) & \longrightarrow & 0 & &
\end{array}$$

Since the bottom left map is $T_n(u_n) = 0$, by exactness $\sigma_{A_n}^T$ is injective, so by a diagram chase we get a unique map $\varphi_n(A) : S_n(A) \rightarrow T_n(A)$ commuting with the σ s.

Now we need to show naturality. Consider a map $f : A' \rightarrow A$.